

Looking Ahead in Goal-Oriented Dialogue

Preference-Tree (PTO) vs Group-Relative (GRPO) optimization of a small Motivational-Interviewing therapist

Results snapshot — Exp3

Numbers and figures only; interpretation deliberately left for the meeting.

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Llama-3.2-1B therapist (bf16) · gpt-4o-mini simulated patient + oracle · 96 personas, persona-paired

Reminder — what we published (ICLR 2025)

Preference Tree Optimization: Enhancing Goal-Oriented Dialogue with Look-Ahead Simulations

Baruch, Butman, Bar, Friedman · ICLR 2025 · attached to this email

- **The method.** At each therapist turn, branch several candidate replies; for each candidate simulate **K further turns** of the conversation; let the oracle score the resulting trajectory; keep the best and worst as a preference pair; update with DPO. Repeat, regenerating the tree from the improved model each round.
- **Why look ahead.** Scoring a reply on its own rewards openings that look good in isolation. Scoring the reply plus K simulated turns rewards openings that lead somewhere.
- **Setup.** Llama-2-7B therapist, GPT-3.5 as both simulated patient and oracle, 96 patient personas, $K \in \{0, 5\}$, 7 iterations, reward = mean(Q1, Q2).
- **Headline finding.** Every PTO model beat the untrained baseline on session satisfaction and working alliance, and **K = 5 gave higher and more stable scores than K = 0.**

What changed since the paper

	Exp1 — ICLR paper	Exp2	Exp3 — this snapshot
Therapist	Llama-2-7B	Llama-3.2-1B, 4-bit	Llama-3.2-1B, bf16
Patient + oracle	GPT-3.5	gpt-4o-mini	gpt-4o-mini
Patient personas	cooperative	less cooperative	less cooperative
Oracle output	regex-parsed	JSON schema	JSON schema
Evaluation	Q1, Q2	6 questionnaires	8 metrics (see next slide)
Methods	PTO	PTO (4 oracles) + weak GRPO	PTO vs GRPO, matched
Iterations	7	7 per arm	10 per arm (K=0)

Each experiment is a fresh re-implementation — no data carries over between them.

One caveat worth stating up front — Exp2 and Exp3 absolute scores are not on the same axis.

Same therapist model, but Exp2 generated its conversations in 4-bit and Exp3 in bf16. 4-bit produces far more degenerate, looping therapist turns ($\approx 9.5\%$ of turns vs $\approx 0.3\%$), which the oracle floors — so Exp2's base model scores 2.38 and Exp3's scores 3.00 for the same model. The non-degenerate Exp2 subset scores ≈ 2.93 . Compare within an experiment, not across.

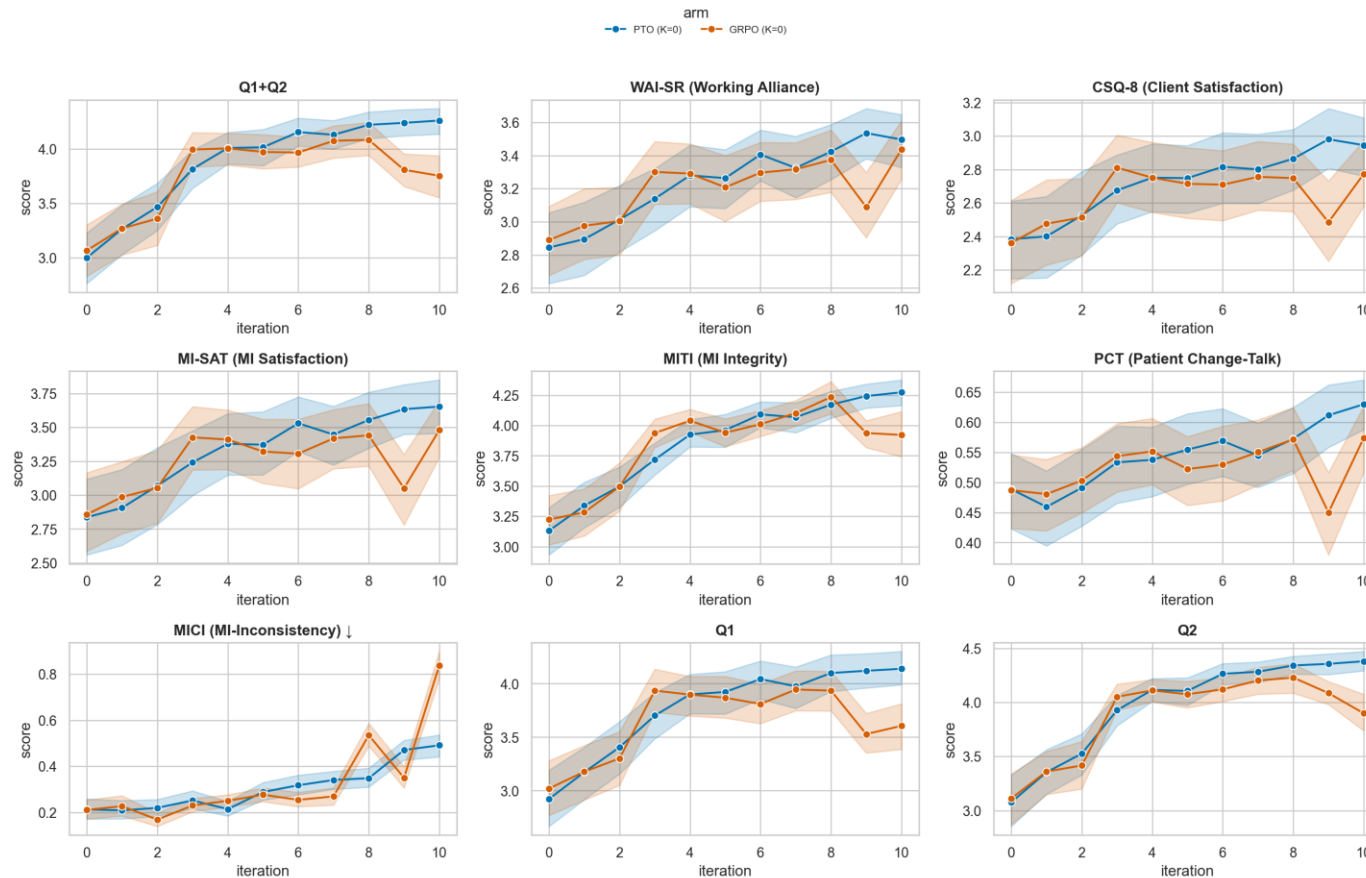
What was run in this experiment

- **Task.** A 1B therapist model converses with a simulated patient (96 fixed personas). A larger oracle model then grades the full conversation on validated MI questionnaires.
- **Training signal.** Oracle score on Q1+Q2 only. All other questionnaires are held out for evaluation.
- **Methods compared.** **PTO** (branch each therapist turn → oracle → preference pairs → DPO update) vs **GRPO** (group-relative update over the same generations). Matched hyper-parameters, matched look-ahead K, matched min-conversation-length filter.
- **Loop.** Each iteration regenerates all conversations from the current policy; those same conversations are the evaluation set.
- **Evaluation.** Full-conversation scores on eight metrics: the five questionnaire rubrics **Q1+Q2**, **WAI-SR**, **CSQ-8**, **MI-SAT**, **MITI**, plus **PCT** (patient change-talk), **MICI** (MI-inconsistent therapist behaviour, lower is better), and the MITI ratios **R:Q / %CR / %MICO** derived from the behaviour counts.

Arm	Iterations trained	Iterations scored	Status
PTO K=0	1–10	base + 1–10	complete
GRPO K=0	1–10	base + 1–10	complete
PTO K=5	1–5	base + 1–4	paused (iter-5 adapter unscored)
GRPO K=5	1	base + 1	paused

All metrics across 10 iterations (K = 0)

Outcome trajectories across iterations — full-conversation eval



The 9 metrics share one dominant global-evaluation (halo) factor (PC1=55%; technique + MI-inconsistency load a second factor; PC2=16%), so a uniform rise partly reflects one axis, not independent multi-skill gains.

Mean over the 96 personas at each iteration; shaded band = 95% CI. Iteration 0 = untrained base model. MICI is inverted in meaning (lower = fewer MI-inconsistent therapist behaviours).

Endpoint numbers (K = 0)

Two endpoints reported for each arm: the matched final iteration (10) and each arm's own best iteration on its training rubric (Q1+Q2).

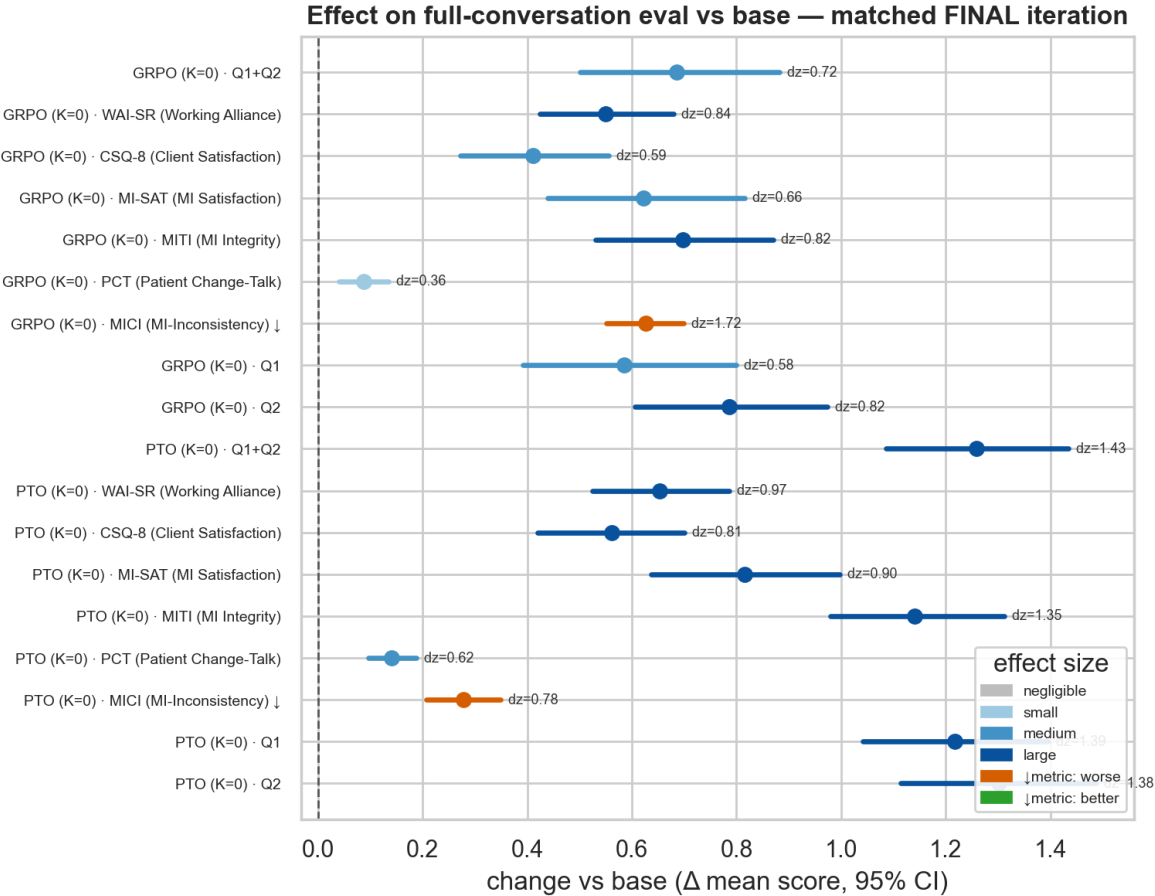
endpoint	arm	iter	Q1+Q2	WAI-SR	CSQ-8	MI-SAT	MITI	PCT	MICI ↓	R:Q	%CR	%MICO
final	PTO (K=0)	10	4.26	3.497	2.945	3.653	4.273	0.63	0.491	0.75	0.364	0.7
final	GRPO (K=0)	10	3.753	3.438	2.773	3.479	3.922	0.574	0.838	1.435	0.414	0.833
best	PTO (K=0)	10	4.26	3.497	2.945	3.653	4.273	0.63	0.491	0.75	0.364	0.7
best	GRPO (K=0)	8	4.082	3.374	2.749	3.441	4.234	0.572	0.535	1.043	0.372	0.711

Paired contrasts on the 96 shared personas (PTO – GRPO), Holm-corrected

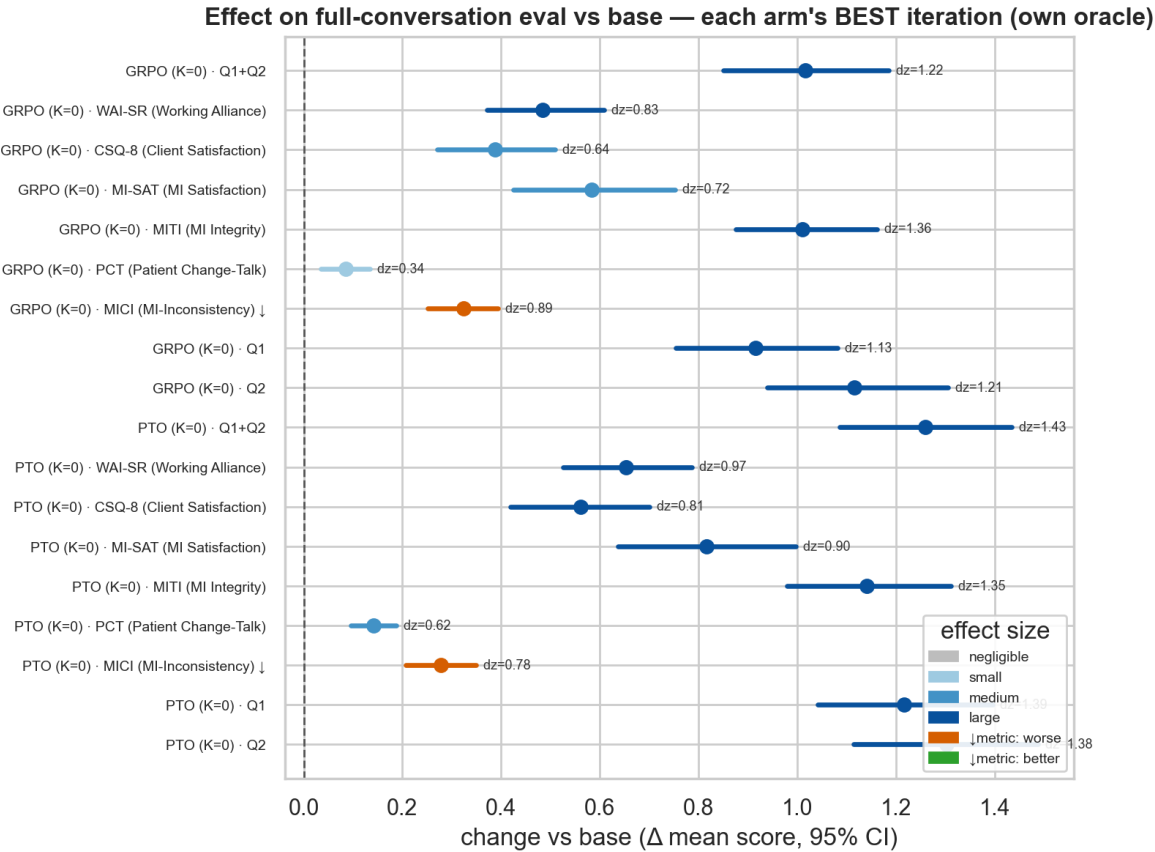
Contrast	Q1+Q2 Δ	dz	p (Holm)
PTO iter-10 vs GRPO iter-10 (matched final)	+0.51	0.73	< .001
PTO iter-10 vs GRPO iter-8 (each at its own best)	+0.18	0.30	.010

Trajectory shape: PTO Q1+Q2 rises 3.00 → 4.26 with its maximum at the last iteration (OLS slope 0.120/iter). GRPO rises 3.07 → 4.08 by iteration 8, then falls to 3.75 by iteration 10 (slope 0.072/iter). Every arm × rubric effect vs base is significant at Holm $p < .001$.

Effect vs the untrained base model (K = 0)



at the matched final iteration (10)

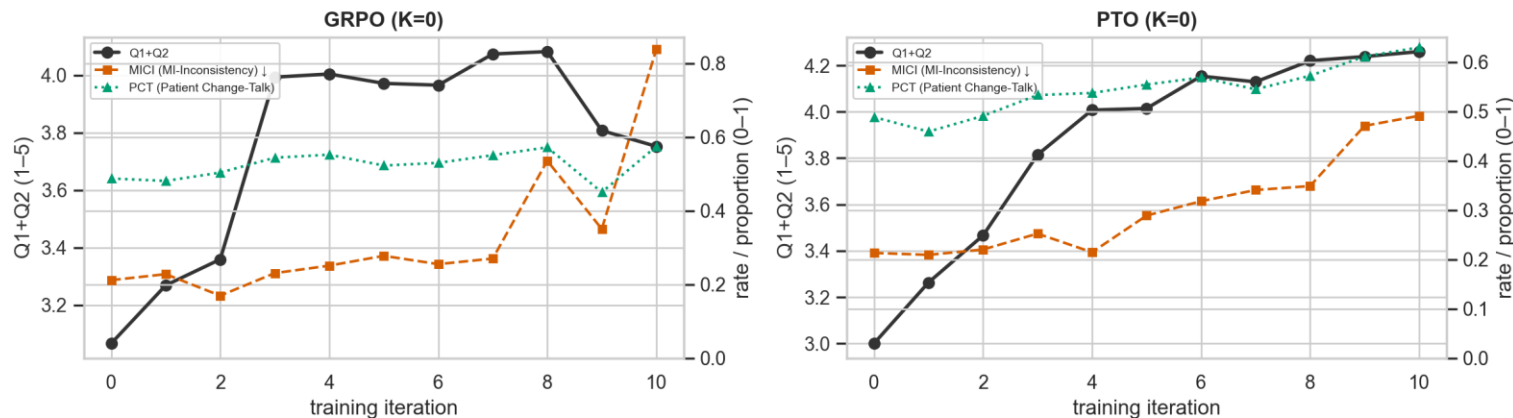


at each arm's own best iteration (PTO 10, GRPO 8)

Paired Cohen's dz vs base over the 96 personas, with 95% CI. Positive = improvement, except MICI where positive = more MI-inconsistent behaviour.

What else moved as the scores rose (K = 0)

Reward-hacking: global-eval reward ↑ and MI-inconsistency ↑ together, patient change-talk ~flat



Left axis = the Q1+Q2 reward proxy (global evaluation; higher = better). Right axis: MI-Inconsistency (red, higher = worse) · Patient Change-Talk (green, higher = better — the actual MI goal).

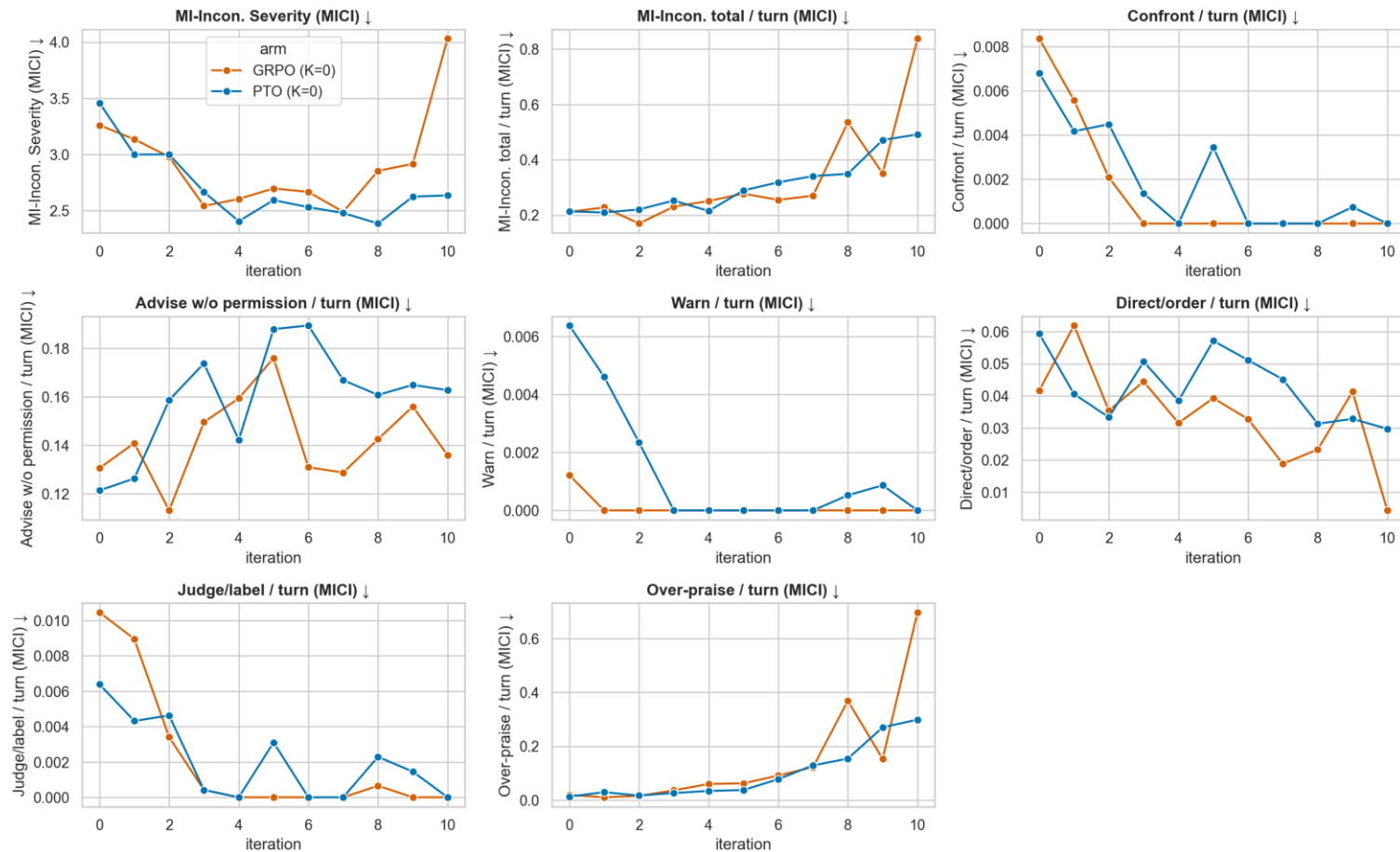
Global rubric score (left axis) vs MI-inconsistent behaviour and patient change-talk (right axis), per iteration

Measured alongside the gains

- **MICI** (MI-inconsistent behaviour) rises 0.21 at base → **0.49** PTO and **0.84** GRPO at iter 10 (GRPO 0.54 at its iter-8 peak).
- Reflection-to-question ratio at iter 10: PTO 0.75, GRPO 1.44.
- The 5 questionnaire rubrics load on a single factor (PC1 ≈ 91%). Adding the other three metrics drops PC1 to ≈ 55% — but that separation comes from **MICI and the MITI ratios**. PCT correlates with the 5 rubrics (ρ 0.79–0.94), so it is not measuring something they miss.
- Both arms drift toward affirmation-heavy language, more so in GRPO's late iterations.
- Full per-item decomposition of all eight scored instruments is on disk (not shown here).

MI-inconsistency, decomposed (K = 0)

MICI detail — severity + per-therapist-turn rates (higher = worse)

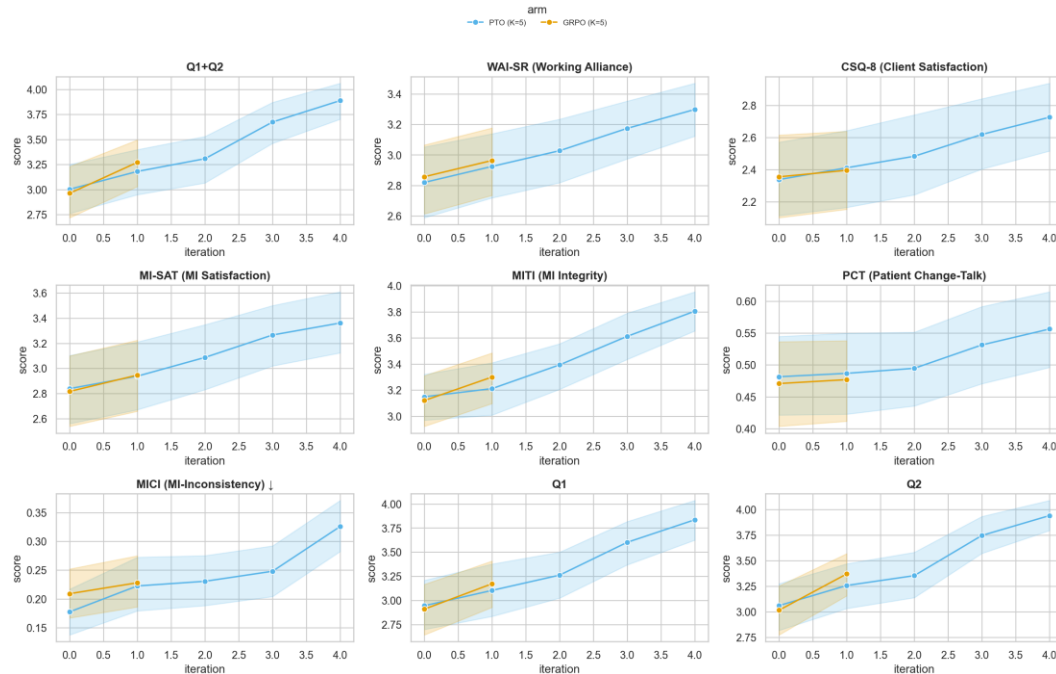


Per-conversation rate of each MI-inconsistent therapist behaviour, by iteration. Lower is better throughout.

Look-ahead (K = 5): exactly what exists today

Arm	Trained	Scored	Missing
PTO K=5	iters 1–5	base + iters 1–4	iter-5 eval conversations never generated
GRPO K=5	iter 1	base + iter 1	iters 2–10

Outcome trajectories across iterations — full-conversation eval



K=5 trajectories — 4 iterations for PTO, 1 for GRPO. Read as preliminary.

Numbers so far (PTO K=5 only)

- Q1+Q2 3.00 → 3.89 over 4 iterations (dz 0.88).
- MICI 0.18 → 0.33 over the same 4 iterations.
- K=0 vs K=5, paired at matched early iterations (PTO): no significant difference on any metric (all Holm $p > .5$).
- GRPO K=5 has one trained iteration — not comparable.
- Both K=5 arms were paused on API cost, not on any result.

Against the ICLR paper: there K=5 beat K=0 on Llama-2-7B over 7 iterations. Here the K comparison rests on 4 iterations of one method and shows nothing yet.

What exists behind these slides

- **2,784 scored conversations** across 29 model checkpoints; every conversation graded on all eight instruments by the oracle.
- **Per-candidate generation records** for every training iteration — all M/G candidates with their scores and sub-scores.
- **Three analysis views** (K=0 only, K=5 only, combined), each with figures, tables and a written summary, regenerated from one command.
- **Per-item decomposition** of every instrument — which individual items move, at both endpoints — already computed, no further oracle calls needed.
- **Persona-level heterogeneity** splits (patient cooperation level, problem type) for every metric.
- **Second-judge reliability pipeline** is built and ready to run (inter-rater agreement + a pluggable second grader model); not yet executed.